

A full-core coupled neutronic/thermal-hydraulic code for the modeling of lead-cooled nuclear fast reactors

R. Bonifetto, S. Dulla, P. Ravetto, L. Savoldi Richard, R. Zanino*

Dipartimento Energia, Politecnico di Torino, 10129 Torino, Italy

HIGHLIGHTS

- First neutronic/thermal-hydraulic full-core modeling of lead-cooled fast reactor with hexagonal fuel elements.
- Separation of time scales allows fast 3D problem solution by 1D + 2D models.
- Space and time numerical convergence demonstrated.
- Accuracy demonstrated by benchmark against analytical solutions.

ARTICLE INFO

Article history:

Received 25 October 2012

Received in revised form 8 March 2013

Accepted 10 March 2013

ABSTRACT

A new multi-physics simulation tool – FRENETIC (Fast REactor NEutronics/Thermal-hydraulics) – is presented for the quasi-3D analysis of a lead-cooled fast reactor core with the hexagonal fuel element configuration, as currently proposed within the framework of the European project LEADER. The tool implements coupled neutronic (NE) and thermal-hydraulic (TH) models. In the NE module, a 2D + 1D full-core multi-group diffusion solver has been developed based on a coarse-mesh nodal scheme and adapted to cope with the hexagonal geometry. In the TH module, the hexagonal elements, described by 1D (axial) transient advection and conduction in the coolant coupled to conduction in the fuel pins, are thermally coupled to each other in the transverse directions to obtain the full-core evolution of the distribution of the TH variables. The NE and TH modules are coupled at each TH time step by transferring to the TH module the distribution of the power source computed by the NE module, which is the driver of the TH evolution; alternately, the temperature distribution computed by the TH module is input to the NE module, in order to update the cross sections. The code is benchmarked against pure TH and pure NE analytical solutions and the results of a coupled NE/TH pseudo-transient (criticality search) are also presented. The convergence of the numerical solution is demonstrated both in space and time by computational simulations.

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

The European Union (EU) sees at present the sodium-cooled fast reactor as the reference choice among Generation IV reactor designs (GIF, 2002), while the lead-cooled fast reactor (LFR) and the gas-cooled fast reactor are being considered as alternative technologies (Deffrennes, 2012).

In the LFR line, on which we shall focus in the present paper, a first design, ELSY (European Lead-cooled System) (Sobolev et al., 2007; Alemberti et al., 2011), was proposed within the 6th European Framework Program (FP6) (<http://www.leader-fp7.eu>, 2012); that design is now being further developed within FP7 under the

Lead-cooled European Advanced DEMonstration Reactor (LEADER) project (Hursin et al., 2012), with the aim of contributing to the design of the Advanced Lead Fast Reactor European Demonstrator (ALFRED) (Alemberti, 2012), toward the full-scale first-of-a-kind European Lead Fast Reactor (ELFR) (Mansani, 2012).

The typical configuration of an LFR core cross section with hexagonal fuel elements is shown in Fig. 1.

The analysis of fast lead-cooled nuclear reactors (Generation IV) requires the use of multi-physics models. In this paper we present a new simulation tool, FRENETIC (Fast REactor NEutronics/Thermal-hydraulics), for the quasi-3D analysis of an LFR core with the hexagonal fuel element configuration.

Without aiming at being comprehensive, one can qualitatively sketch the collocation of FRENETIC in the landscape of already available computational tools as in Fig. 2. We distinguish between purely TH codes, purely NE codes and coupled NE/TH codes: in the first group we find ATHENA (Davis and Shieh, 2000) and FEM-LCORE

* Corresponding author at: Dipartimento Energia, Politecnico di Torino, c. Duca degli Abruzzi 24, 10129 Torino, Italy. Tel.: +39 011 090 4490; fax: +39 011 090 4499.
E-mail address: roberto.zanino@polito.it (R. Zanino).

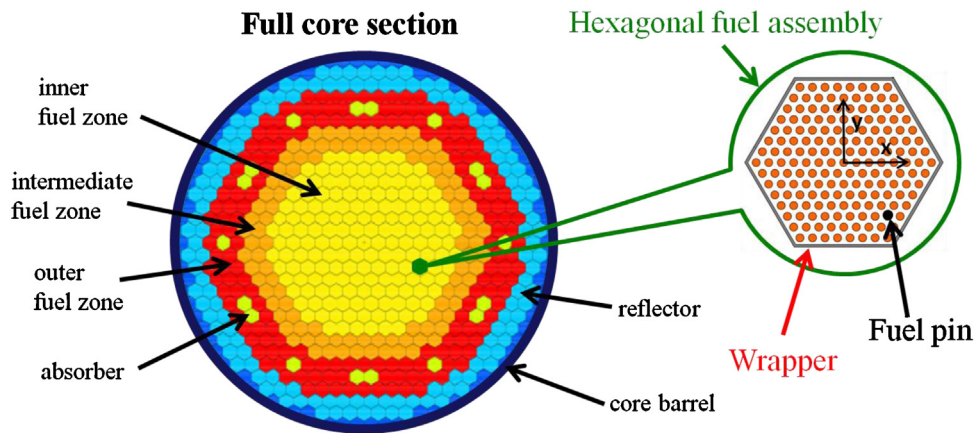


Fig. 1. Sketch of the ELFR core cross section. Detail of the hexagonal fuel assembly.

Reproduced from Alemberti et al. (2011).

(Mikhin et al., 2011); in the second group we find TORT (Rhoades and Azmy, 1995), ERANOS (Rimpault et al., 2002), APOLLO (Sanchez et al., 2010) and MCNP (www.lanl.gov, 2012), this list including in turn codes of different types (deterministic vs. statistic, steady state vs. transient, etc.); in the third category we find RELAP/PARCS (www.inl.gov, 2012; Hursin et al., 2012; Meloni et al., 2008), SIMMER (Yamano et al., 2008), FAST (Mikityuk et al., 2005), CATHARE (www.cathare.cea.fr) and FRENETIC. In Fig. 2 we also distinguish between codes (as FRENETIC) applicable to the full core, or to the whole reactor (i.e., including also the external circuit(s) feeding the core).

In order to understand the added value of FRENETIC with respect to the landscape of already available tools, we summarize in Table 1 the main features of those tools which, as FRENETIC, include a coupled neutronic/thermal-hydraulic model.

It is seen in Table 1 that FRENETIC is the only tool providing a coupled NE/TH description in hexagonal geometry at the full-core level.

The model assumes a uniform distribution of the different (TH and NE) relevant parameters at the hexagon level. Therefore, while

it cannot treat transients at the so-called sub-assembly level (e.g., severe transients leading to loss of structural integrity of single pins or groups of pins, etc.), this assumption is well justified for several operational and accidental transients of interest.

As the name says, the code has the ambition to provide computationally effective (i.e., relatively fast) approximate solutions for core design and/or safety analysis, thanks to the fact that the 3D problem is solved with a simplified approach.

In a first step of this development, we recently coupled a TH model for a single hexagonal element to a point-kinetic NE model (Bonifetto et al., 2012a). The objective of the present work is the extension of that analysis to the full core configuration. In the NE module, a multi-group diffusion solver is implemented for the determination of the neutron distribution based on a coarse-mesh nodal scheme, adapted to cope with the hexagonal geometry. In the TH module, the full core is treated as a set of hexagonal elements thermally coupled to their neighbors. The coupling between the TH and the NE modules is performed explicitly using the commercial TISC[®] platform (www.tlk-thermo.com, 2012). Although, for the time being, the NE model is steady state, this solution will allow

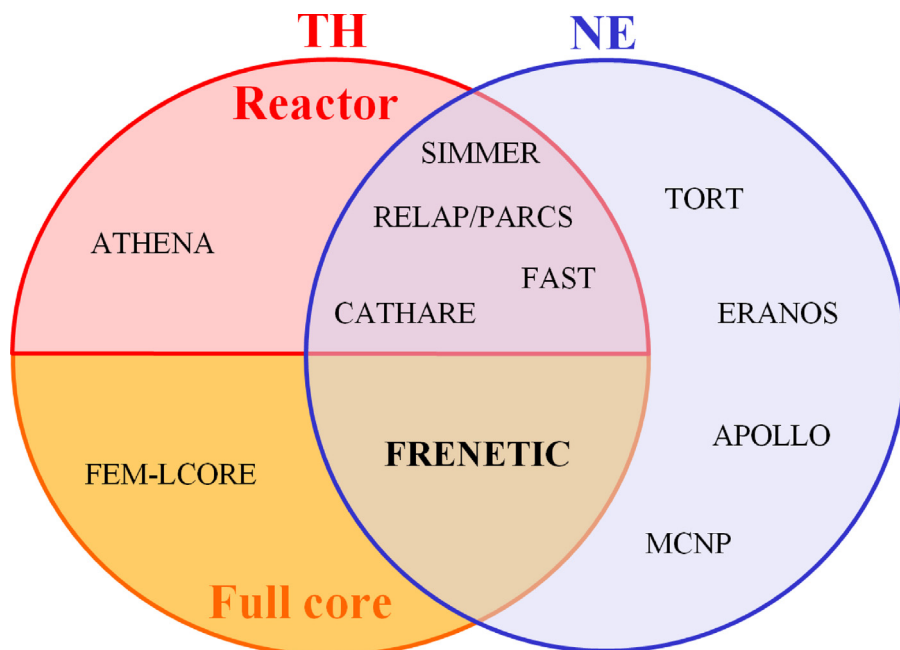


Fig. 2. A sketchy representation of available tools for nuclear reactor neutronic/thermal-hydraulic modeling.

Table 1

Main features of the FRENETIC code compared to the already available tools for coupled neutronic/thermal-hydraulic modeling of nuclear reactors.

		SIMMER IV	RELAP5 + PARCS	RELAP5-3D (+NESTLE)	FAST (TRACS/AAA + PARCS)	CATHARE	FRENETIC
Neutronic analysis	Transport theory	×				×	
	Diffusion approximation		×	×	×		×
	Dimension	3D	3D	1D–2D–3D	3D	3D	2D→3D
NE geometry	Cartesian/cylindrical	×	×	×	×	×	
	Hexagonal		×	×	×		×
Thermal hydraulic analysis	Fuel assembly	×					×
	Core	×	×	×	×	×	×
	System	×	×	×	×	×	
	Original working fluid	Na	Water	Water	He	Water	Pb & LBE
	Dimension	3D	2D	3D	3D	1D–3D	1D + 2D
TH geometry	Cartesian/cylindrical	×	×	×	×	×	
	Hexagonal						×
Mechanical analysis		×			×	×	

high flexibility in terms of using different internal time steps for the two modules, according to the different NE and TH time scales; furthermore, this coupling strategy is very fast if compared with I/O on data files, and it allows to keep two separate main codes (NE and TH) that can be, as such, also independently run to separately test and validate them. The test of the two main codes, including the successful comparison with analytical solutions in simplified cases, has been done in the frame of the present work and constitutes an important step for future developments. Furthermore, the TH module was also successfully validated in the meanwhile against experimental results at the level of a single hexagonal channel (Zanino et al., 2012; Bonifetto et al., 2012b). The work presented here also proves the effectiveness of the coupling strategy between the two modules in FRENETIC, allowing an efficient and correct data exchange.

In conclusion, already in the present form the newly developed code can be fruitfully used for the static design and optimization of a lead-cooled system at full power. The NE spatial-spectral solution together with the full TH calculation is by far more accurate than lumped parameter models (Bortot et al., 2012) for the engineering evaluation of lead-cooled systems.

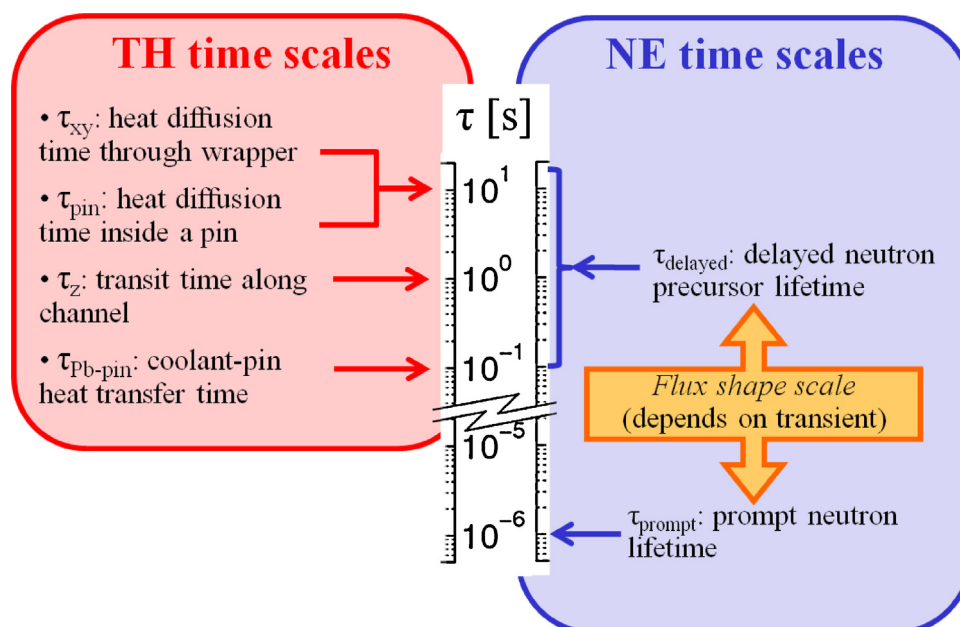
2. Model description

The model presented in this section and implemented in FRENETIC aims at a quasi-3D description of the LFR core. This is achieved by coupling a transient “1D+2D” TH model to a steady-state “2D+1D” NE model. The meaning of the above definitions is as follows: the TH model consists of 1D axial/vertical models of each hexagonal fuel assembly, thermally coupled on the horizontal plane (2D); the NE model (for the time being) provides a steady-state 2D description of the neutron flux on the horizontal plane, while a cosine shape is assumed in the axial/vertical direction (1D). In order to justify this approach to the fully 3D original problem we consider in the next section the different time scales of interest.

2.1. Relevant time scales

Several different and to some extent separate time scales are relevant for the multi-physics problem of the LFR dynamics. These are summarized in Fig. 3, distinguishing between TH and NE phenomena.

The time scales in Fig. 3 are defined and have been estimated as follows:

**Fig. 3.** Relevant time scales for LFR dynamics.

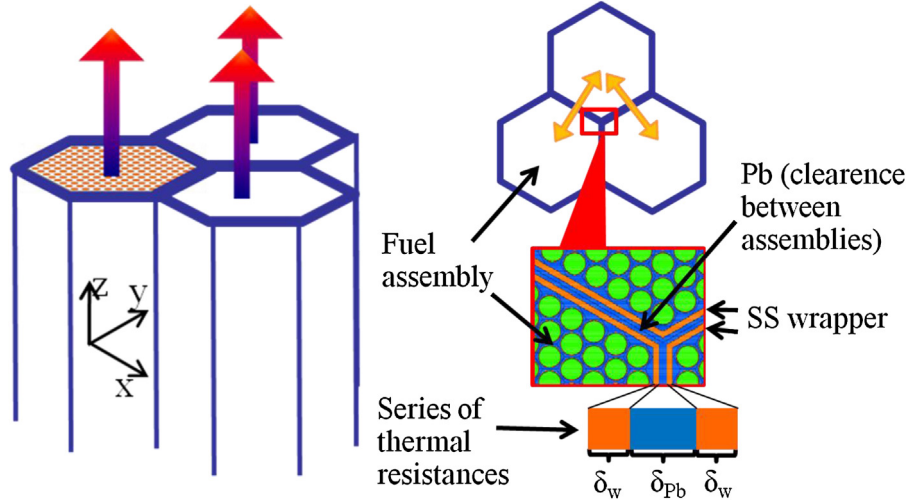


Fig. 4. Core schematics adopted for the TH model.

- $\tau_{xy} \approx \delta^2/\alpha$, where $\delta = \delta_{pb} + 2 \times \delta_w$ is the total thickness separating two neighboring hexagons, including the Pb clearance (see Fig. 4) where the Pb flow is considered negligible (the Pb is assumed to be stagnant) and $\alpha = (\alpha_{pb} \times \delta_{pb} + 2 \times \alpha_w \times \delta_w)/\delta$ is the average (weighted) thermal diffusivity including the effect of the three layers;
- $\tau_{pin} \approx r_{pin}^2/\alpha_f$, where r_{pin} is the pin radius and α_f is the fuel thermal diffusivity (the presence of the cladding is neglected for the time being);
- $\tau_z \approx H/V$, where H is the active core height (i.e., the length of the portion of pin containing the fuel) and V is the coolant speed in the axial direction;
- $\tau_{pb-pin} \approx (A_{pb} \times c_v \times \rho_{pb})/(h \times \Pi_f)$, where A_{pb} is the Pb flow area, c_v is the Pb specific heat at constant volume, ρ_{pb} is the Pb density, h is the heat transfer coefficient between the coolant and the pin surface, Π_f is the fuel wetted perimeter.

2.2. Thermal-hydraulic model

The TH model in FRENETIC takes advantage of the separation of time scales $\tau_{xy} \gg \tau_z$. Since the thermal coupling between neighboring hexagons is relatively slow/weak compared to the Pb transit time along the fuel assembly, the 3D problem can be approximated as a set of weakly (i.e., explicitly) coupled 1D problems, to be solved along each fuel assembly. This approach is very similar to what we previously introduced and successfully used in the context of cryogenic cooling of superconducting magnets for nuclear fusion reactors (Savoldi and Zanino, 2000).

The TH module of FRENETIC solves implicitly in time (fully implicit or Crank–Nicolson schemes) the transient 1D mass, momentum and energy balances in each fuel assembly for the z component $V(z, t)$ of the flow velocity, and for the temperature $T_{pb}(z, t)$ and pressure $p(z, t)$ pair.

Concerning the fuel temperature, two assumptions are made: (a) a single, suitably averaged temperature is assumed to be representative of the temperature profile inside each pin; (b) all pins inside a hexagon are assumed to be described by the same fuel temperature $T_f(z, t)$. The latter assumption is justified, as seen above, by the uniform distribution of the fission heat source among the pins of any given fuel assembly and provided boundary effects are negligible at the hexagon level.

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial z} + \frac{1}{\rho_{pb}} \frac{\partial p}{\partial z} = -FV - g \quad (1)$$

$$\begin{aligned} \frac{\partial p}{\partial t} + \rho_{pb} c_s^2 \frac{\partial V}{\partial z} + V \frac{\partial p}{\partial z} - G \frac{\partial}{\partial z} \left(k_{pb} \frac{\partial T_{pb}}{\partial z} \right) + \rho_{pb} c_s^2 \frac{V}{A_{pb}} \frac{\partial A_{pb}}{\partial z} \\ = G \left[\frac{\Pi_f h^*}{A_{pb}} (T_f - T_{pb}) + V \rho_{pb} F \right], \end{aligned} \quad (2)$$

$$\begin{aligned} \rho_{pb} c_v \frac{\partial T_{pb}}{\partial t} + \rho_{pb} c_v V \frac{\partial T_{pb}}{\partial z} + \rho_{pb} c_v G T_{pb} \frac{\partial V}{\partial z} - \frac{\partial}{\partial z} \left(k_{pb} \frac{\partial T_{pb}}{\partial z} \right) \\ + \rho_{pb} c_v G T_{pb} \frac{V}{A_{pb}} \frac{\partial A_{pb}}{\partial z} = \frac{\Pi_f h^*}{A_{pb}} (T_f - T_{pb}) + V \rho_{pb} F + \sum_{i=1}^6 \frac{Q_i}{A_{pb}}, \end{aligned} \quad (3)$$

$$\rho_f c_f \frac{\partial T_f}{\partial t} - \frac{\partial}{\partial z} \left(k_f \frac{\partial T_f}{\partial z} \right) = \frac{\Pi_f h^*}{A_f} (T_{pb} - T_f) + \frac{Q_f}{A_f}, \quad (4)$$

where A_f is the fuel cross sections, c_f is the fuel specific heat, c_s is the sound speed, g is the gravity acceleration, h^* is a heat transfer coefficient between fuel and coolant described below, k_{pb} and k_f are the coolant and fuel thermal conductivities, respectively, p is the Pb pressure, Q_i is the heat exchanged between the currently considered hexagon and the i -th neighboring hexagon, see below, Q_f is the linear power generation in the fuel, see below, T_{pb} and T_f are the coolant and fuel temperature, respectively, z is the axial coordinate, ρ_f is the fuel density.

In (1)–(4), F , accounting for friction effects, and G , the Gruneisen parameter, are defined as follows

$$F = 2f \frac{|V|}{D_h} \quad (5)$$

$$G = \left(\frac{\rho_{pb}}{T_{pb}} \frac{\partial T_{pb}}{\partial \rho_{pb}} \right)_s \quad (6)$$

where f is the Fanning friction factor, s is the entropy and D_h is the hydraulic diameter.

In (2)–(4), the heat transfer term $h \times [T_s(z, t) - T_{pb}(z, t)]$ between the pin surface at temperature T_s and the Pb, occurring on a time scale τ_{pb-pin} defined by the heat transfer coefficient h , taken here from the Schad modified correlation for pin bundles, see chapter 10 of (Todreas and Kazimi, 1993), is implemented as $h^* \times [T_f(z, t) - T_{pb}(z, t)]$, where

$$h^* = h \times \frac{T_s(z, t) - T_{pb}(z, t)}{T_f(z, t) - T_{pb}(z, t)} \quad (7)$$

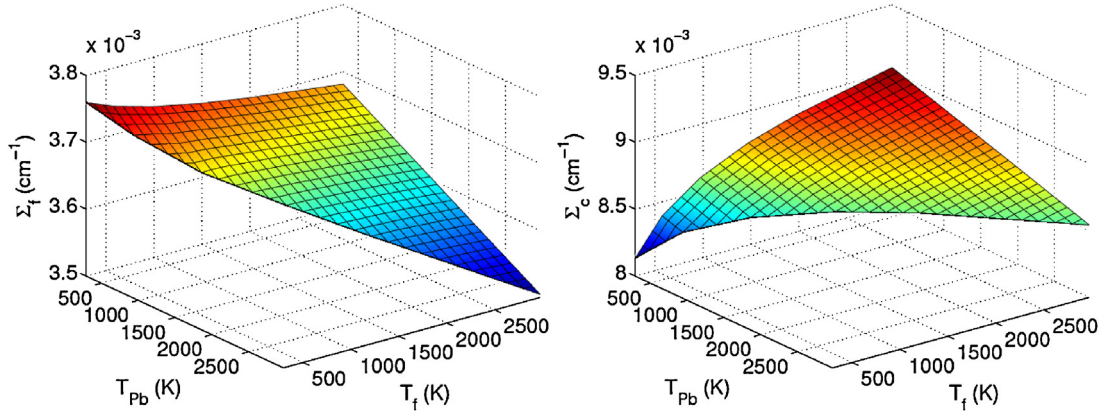


Fig. 5. Behavior of fission and capture cross sections for the third energy group as a function of fuel and Pb temperature (Carta, 2012), as assumed in the present paper.

and use in h^* an approximate value for T_s , estimated assuming a parabolic radial temperature distribution inside the pin (since this is modified only on the time scale $\tau_{\text{pin}} \gg \tau_{\text{Pb-pin}}$). Then T_s can be analytically linked to the heat generation rate $q_v(z, t) = Q_f/A_f$ and to $T_f(z, t)$ as follows:

$$T_s(z, t) = \frac{T_f(z, t) - q_v(z, t) \times r_{\text{pin}}^2}{8 \times k_f}. \quad (8)$$

In (1)–(4), space derivatives along z are approximated by 1D linear finite elements, equivalent to central difference approximation. The resulting set of ODEs is then marched forward in time using the Crank–Nicolson scheme.

The TH model must be coupled to an NE model because the main driver in the TH evolution is the fission heat source inside the fuel

$$Q_f = A_f \cdot E_f \cdot \sum_{g=1}^G \Sigma_{f,g} \cdot \Phi_g, \quad (9)$$

where A_f is the (total) pin cross section, E_f is the energy produced per fission, $\Sigma_{f,g}$ is the macroscopic fission cross section of group g , Φ_g is the neutron flux value for group g and G is the number of energy groups.

In the 1D Pb balances (1)–(3), thermal coupling terms appear with neighboring hexagons which are treated explicitly in time since they are weak. This heat transfer is given by

$$Q_i = \frac{\Pi_{ij} h_{ij}}{A_{\text{Pb}}} (T_{\text{Pb},i} - T_{\text{Pb},j}), \quad (10)$$

where Π_{ij} is the contact perimeter between hexagon i and hexagon j , h_{ij} is the heat transfer coefficient,

$$h_{ij} = \left(2 \frac{1}{\alpha_w} + \frac{1}{\alpha_{\text{pb}}} \right)^{-1} \quad (11)$$

computed from the series of thermal resistances (SS wrapping–Pb clearance–SS wrapping) between hexagon i and hexagon j , as shown in Fig. 4 (SS = stainless steel). In the clearance, the Pb is assumed to be stagnant, so its contribution to the heat transfer coefficient h_{ij} is of purely conductive nature.

2.3. Neutronic model

The description of the neutron flux in FRENETIC is based, for the time being, on the multi-group diffusion model

$$\begin{aligned} \nabla \cdot D_g(T_f, T_{\text{Pb}}) \nabla \Phi_g - \Sigma_{r,g}(T_f, T_{\text{Pb}}) \Phi_g + \sum_{g'=1}^{g-1} \Sigma_{s,g' \rightarrow g}(T_f, T_{\text{Pb}}) \Phi_{g'} \\ + \frac{1}{k_{\text{eff}}} \chi_g \sum_{g'=1}^G \nu \Sigma_{f,g'}(T_f, T_{\text{Pb}}) \Phi_{g'} = 0 \quad g = 1, \dots, G, \end{aligned} \quad (12)$$

where D is the diffusion coefficient, Σ_r is the removal cross section, $\Sigma_{g' \rightarrow g}$ is the scattering cross section from group g' to group g , k_{eff} is the effective neutron multiplication factor, χ is the fission spectrum, ν is the number of neutrons produced by each fission reaction. Therefore, delayed neutrons are not present in the current implementation of the code and a steady state is assumed on the fastest time scale τ_{prompt} .

In order to perform coupled calculations on a full-core configuration, the energy description of the neutron population needs to be reduced to a small number of groups. We adopt a set of three-group cross sections since this choice allows us to properly describe the behavior of the fast neutron spectrum within the system, constituting thus a good compromise between accuracy and computational cost.

Since both the diffusion coefficient and the different cross sections depend on both fuel and Pb temperature, the NE model needs the temperature profiles computed by the TH model. Among these dependences the most significant ones (and the only ones taken into account in the present paper) are those of the removal and fission cross sections. The cross section database, including their temperature dependence, was generated for a lead-cooled fast system using the ECCO module of the transport code ERANOS (Rimpault et al., 2002), to produce the energy spectra (Carta, 2012). In Fig. 5 the temperature dependence of the fission and capture cross sections for the third energy group ($g=3$) is reported.

While Σ_f decreases both for increasing T_f and for increasing T_{Pb} , giving rise to a negative temperature feedback, the capture cross section $\Sigma_c = \Sigma_r - \Sigma_f$ increases for increasing T_f , due to the increased resonance captures, and decreases for increasing T_{Pb} . The latter behavior, related to the Pb density reduction for increasing temperature, may in turn be responsible of a positive temperature feedback. Although the variation of the cross sections in Fig. 5 is relatively small, for relatively small variation of the temperature, it has to be recalled that this variation is then reflected in a corresponding variation of k_{eff} , which albeit relatively small obviously has important implications.

The NE model is solved in FRENETIC adopting the coarse-mesh approach (2D nodal scheme) (Malambu and Mund, 1995) for the full-core evaluation in hexagonal geometry. Each fuel element

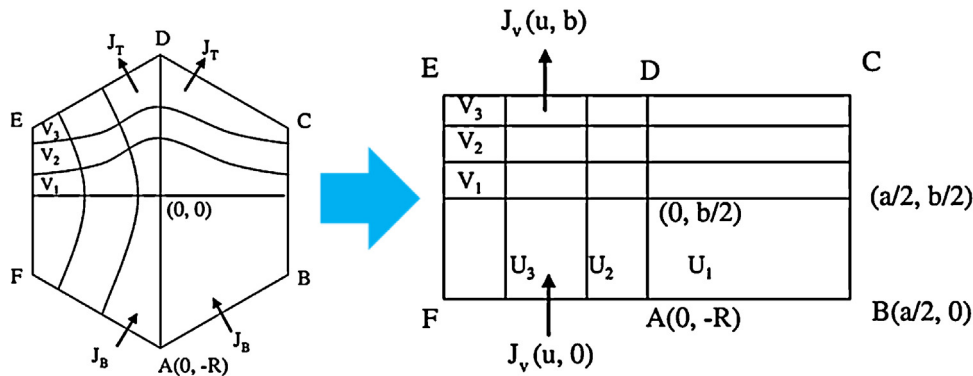


Fig. 6. Hexagonal geometry of the fuel element (left) and its mapping onto a rectangular domain (right).

constitutes a computational node in the x – y plane, while the axial variable needs to be discretized, to generate a 3D grid for the solution of the diffusion equation. The cross sections are supposed to be homogeneous on each hexagon, as it is customarily performed in nodal method, and a third order expansion of the neutron flux within the hexagon is assumed.

The formulation of the nodal method in hexagonal geometry requires a change of coordinates. A 2D conformal mapping is applied, as sketched in Fig. 6. The transverse integration of the balance equations generates a set of equations for the incoming and outgoing currents for the 6 edges of the hexagon, to be coupled to the neighboring fuel elements (Lawrence, 1983).

For the time being, in the vertical z direction a buckling term is introduced to account for axial neutron losses. This corresponds to imposing a cosine shape of the flux along the axis, with an amplitude such that

$$\int_0^H P(z) \cdot dz = P_{\text{hexagon}}, \quad (13)$$

where $P(z)$ is the linear power density distribution along the fuel element and P_{hexagon} is the total power produced by each hexagonal assembly, according to the neutron flux computed in the 2D hexagonal element. The linear power density distribution $P(z)$ is assumed to be

$$P(z) = a + (\varphi - 1)a \cos \left(\frac{\pi}{H}z - \frac{\pi}{2} \right), \quad (14)$$

where φ is the axial power form factor, i.e., the ratio of maximum to minimum linear power density in each pin (assembly in this work) and a is a parameter computed solving (13).

In perspective, the solution of the time-dependent problem for operational and accidental transient analysis will be performed inverting the full space-energy problem on the time scale of the prompt neutrons. This approach may constitute a reference, and is characterized by a larger computational effort. For this reason, we are envisaging the implementation of a quasi static procedure (Dulla et al., 2008), allowing a good accuracy in the reproduction of spatial and spectral effects with a reduced computational time.

2.4. Neutronic/thermal-hydraulic coupling strategy

The coupling between the TH and the NE models takes advantage of the fact that the neutron dynamics is (much) faster than the thermal-hydraulic dynamics, see Fig. 3.

We can then couple the NE and TH modules of FRENETIC explicitly on the shortest thermal-hydraulic time-scale ($\sim \tau_z$), while each module could march forward in time with a time step choice most suitable for the resolution of the respective (fastest) time scale of interest. In the present work, as already observed,

only the steady-state neutron distribution is solved for, taking advantage of the fact that the evolution of the prompt neutrons population (the only included in the model for the time being) occurs on the fastest time scale.

The coupling between TH and NE modules is performed in practice using the commercial TISC platform (www.tlk-thermo.com, 2012), which we already successfully used in the modeling of nuclear fusion problems (Savoldi Richard et al., 2010).

3. Results

The performance of FRENETIC has been tested in a few selected test cases reported here. The successful validation of the TH model in the case of a single hexagon, against experimental data from the ICE test section at the ENEA Brasimone research center, has already been presented elsewhere (Zanino et al., 2012; Bonifetto et al., 2012b).

In all cases the cross section of the computational domain is the same, extending out to (but not including) the reflector region (see Fig. 1). With respect to Fig. 1 we assume that all fuel zones have the same composition and no control absorber is present, for the sake of simplicity. The fuel is assumed to be UO_2 (Popov et al., 2000), while the wrapper material is 18/12 stainless steel (similar to AISI 316).

The number of hexagons (fuel elements) in the fuel region of the reference configuration is 451, unless otherwise noted, the reactor thermal power is 1500 MW and the axial power form factor $\varphi = 1.3$ (Sobolev et al., 2007). The other main geometrical parameters, taken from (Sobolev et al., 2007; Mansani, 2012), are listed in Table 2.

The Pb flows from below into the core, under the action of an imposed pressure gradient Δp (or forced mass flow rate, depending on the case) and of the gravity force. The Pb properties are taken from (OECD-NEA, 2007). The friction factor correlation for pin bundles is taken from chapter 9 of (Todreas and Kazimi, 1993).

The inlet temperature is assumed to be constant and equal to 673 K.

Table 2

Main geometrical characteristic of the ELFR core considered in this paper for the tests of the FRENETIC code.

Barrel diameter (m)	5
Active core length (m)	1.2
Hexagon side length (mm)	120.67
Number of pins/hexagon	169
Pin diameter (mm)	10.5
Fuel pin pitch (center-to-center) (mm)	15
Wrapper thickness (mm)	4
Pb clearance between assemblies thickness (mm)	5

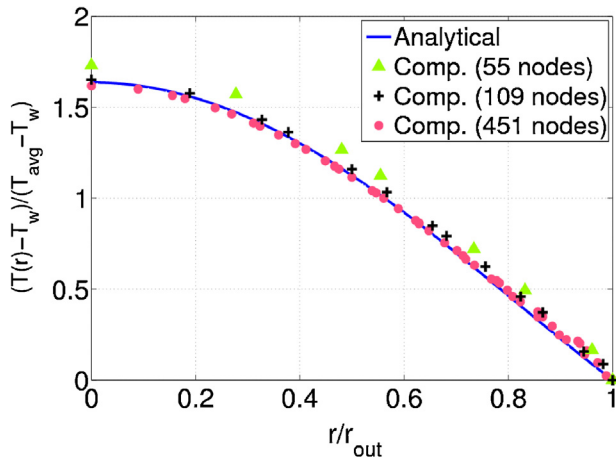


Fig. 7. Comparison between computed and analytical radial profiles of the Pb temperature in a smooth pipe with laminar flow conditions and imposed heat flux on the side walls, for different number of nodes (= hexagons) used to discretize the fixed core cross section area.

3.1. A pure-TH model problem

This first computational test of FRENETIC aims at comparing the 2D Pb temperature distribution computed at the core outlet section with the fully developed temperature profile

$$\frac{T(r) - T_w}{T_{avg} - T_w} = \frac{96}{11} \cdot \left[\frac{3}{16} + \frac{1}{16} \cdot \left(\frac{r}{r_{out}} \right)^4 - \frac{1}{4} \cdot \left(\frac{r}{r_{out}} \right)^2 \right], \quad (15)$$

analytically computed (Incropera and Dewitt, 2006) at the outlet of a straight smooth circular pipe having the same cross section area of the core, in the case of laminar Pb flow and pipe walls subjected to a constant heat flux q_{pipe} . In (15), $T(r)$ is the temperature value at radius r , T_w is the wall temperature, T_{avg} is the average temperature on the cross section, r_{out} is the equivalent cylindrical core radius (2.33 m) and the denominator $T_{avg} - T_w$ is proportional to q_{pipe} . The side boundaries of the core are subjected to the same heat flux q_{pipe} .

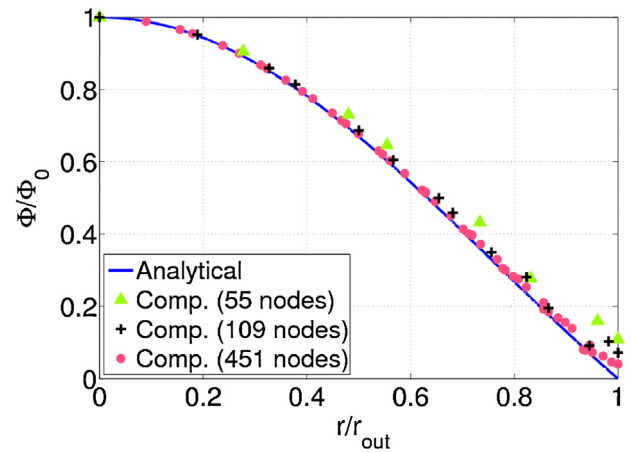


Fig. 8. Comparison between computed and analytical radial profiles of the neutron flux in a cylindrical bare reactor, for different node numbers (hexagons) used to discretize the fixed core cross section area.

The laminar flow assumption justifies the comparison with a model as ours, which assumes no mixing of the flow between neighboring hexagons. In the computational test, the inlet flow velocity imposed as boundary condition in each hexagon is given by the value of the axial velocity component of the fully developed laminar flow profile

$$V(r) = \frac{1}{4\mu} \frac{dp}{dz} (r^2 - r_{out}^2) = 2V_{ave} \left[1 - \left(\frac{r}{r_{out}} \right)^2 \right] \quad (16)$$

over the core cross section area, taken in the center of the hexagon. In (16), μ is the dynamic viscosity and V_{ave} is the average speed.

As the analytical solution (15) applies to a problem without internal boundaries, the boundaries between hexagons are assumed to reduce to a very thin layer of Pb and negligible thickness of the wrapping ($\delta_{pb} = 1$ mm, $\delta_w = 0$).

The computed values of the outlet temperature for each hexagon are compared in Fig. 7 with the analytical result from (15). Although the number of fuel elements (hexagons) is obviously fixed in a

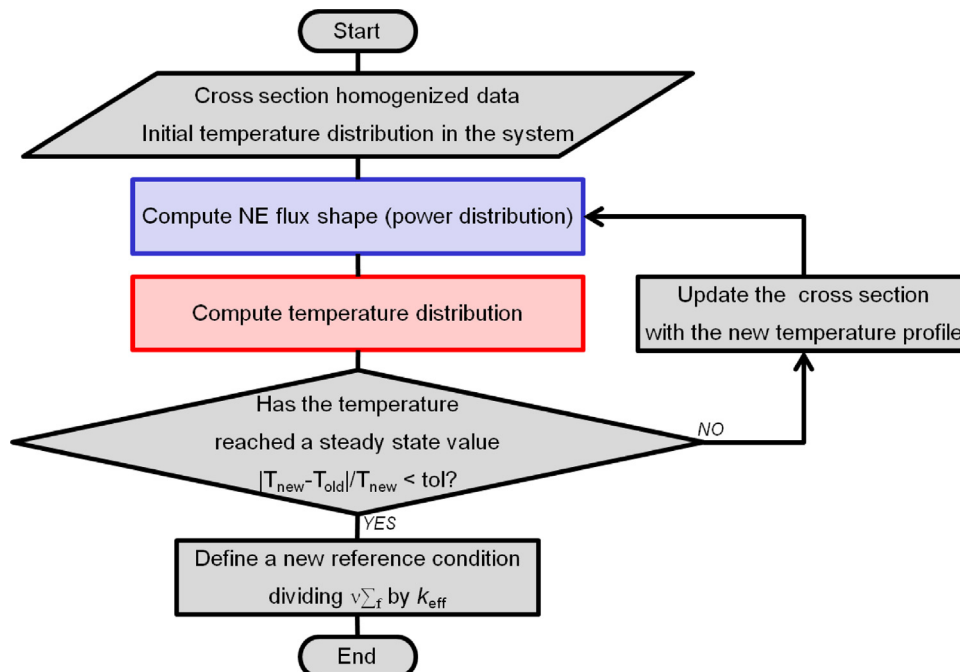


Fig. 9. Procedure for the criticality search.

given reactor design, we treat it in this section simply as the discretization parameter for the 2D problem on the cross section. The number of computed points in Fig. 7 is smaller than the element number because, due to the symmetry of the problem, only values from elements at different radial distances from the core center are reported. As expected, the computed results show an increasingly good agreement with the analytical solution, for increasing number of hexagonal channels used in the discretization of the fixed computational domain. This confirms the adequacy of the heat transfer model between neighboring hexagons implemented in the FRENETIC code.

3.2. A pure-NE model problem

In the second computational test, the 2D distribution of the neutron flux computed by the NE module of FRENETIC is compared with the analytical distribution in the case of a cylindrical bare reactor, which for a single energy group and an isothermal reactor is given by (Abramowitz and Stegun, 1965)

$$\frac{\Phi(r)}{\Phi_0} = J_0 \left(2.4048 \cdot \frac{r}{r_{out}} \right), \quad (17)$$

where Φ_0 is the flux at the center of the core and J_0 is the Bessel function of order zero.

Also in this case the number of hexagons in the domain has been varied (with a fixed core cross section area) treating each hexagon as a “node” of our discretization. Again, the larger the number of nodes, the better the agreement between computed and analytical profiles (see Fig. 8). The small discrepancies at the outer boundary of the domain are related to the somewhat different boundary conditions applied in the two cases: the analytical solution (17) is obtained imposing a zero-flux condition on the boundary, while the computational solution assumes a zero incoming current.

3.3. A coupled TH/NE model problem: criticality search

The third and last computational test presented in this paper aims at assessing the quality of the computed solution in the case of a coupled NE/TH model problem, the so-called criticality search (Fig. 9). For this test case, a Δp of 0.235 MPa is imposed to obtain the desired flow speed of 1.1 m/s at full power.

Before the first coupling between the two modules is performed, a preliminary, purely NE simulation is carried out, aimed at computing k_{eff} in the “cold reactor” condition, i.e., with the whole reactor

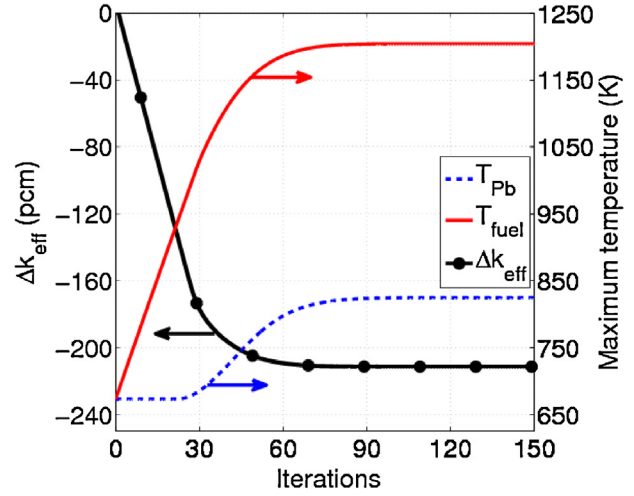


Fig. 10. Behavior of the Δk_{eff} (solid symbols, left axis) and of the maximum temperature of fuel (solid line) and Pb (dashed line) during the criticality search test (the TH time step $\Delta t = 0.1$ s corresponds to one iteration of the NE module, $\Delta z = 0.01$ m).

at the same temperature level (assumed here equal to the Pb inlet temperature). The resulting $k_{eff} = k_{eff}^{cold} = 1.09476$. The cold multiplication factor $k_{eff}^{cold} > 1$ using the set of nuclear data and the geometrical configuration as described above. In a real design, the material composition needs to be adjusted, also introducing control elements, in order to produce a critical hot state. In the present evaluation, however, we shall be mainly interested to compute the amount of reactivity introduced (Δk_{eff}) when going from a cold to the hot situation.

At this point the procedure reported in Fig. 9 is followed: starting again from a uniform temperature in the core (assumed equal to the Pb inlet temperature), the cross sections are evaluated and the neutron fluxes are calculated. The distribution of the power generation from fission reactions is computed from the flux shape, assuming that the power level is the nominal reactor thermal power (1500 MW). The power distribution is then used in the TH module to compute the new temperature distribution. Then, unless the relative temperature variation with respect to previous value is smaller than a given tolerance tol ($= 10^{-6}$ in our case), the value of the different cross sections is updated in the NE module, using the new temperature values, and a new neutron flux shape is computed. In this iterative process, the TH module computes the temperature

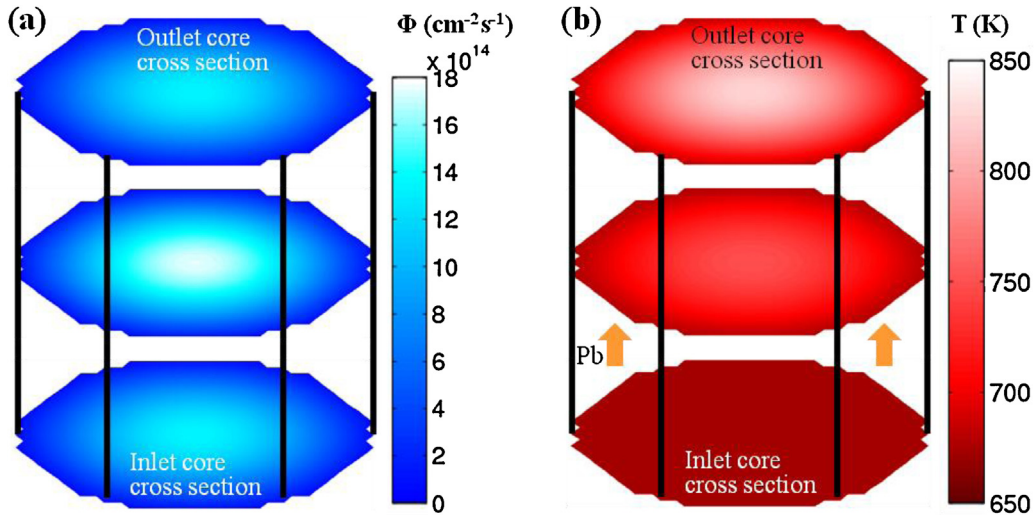


Fig. 11. 2D maps of the fast ($g = 1$) neutron flux (a) and of the Pb temperature (b), at three different axial locations in the core.

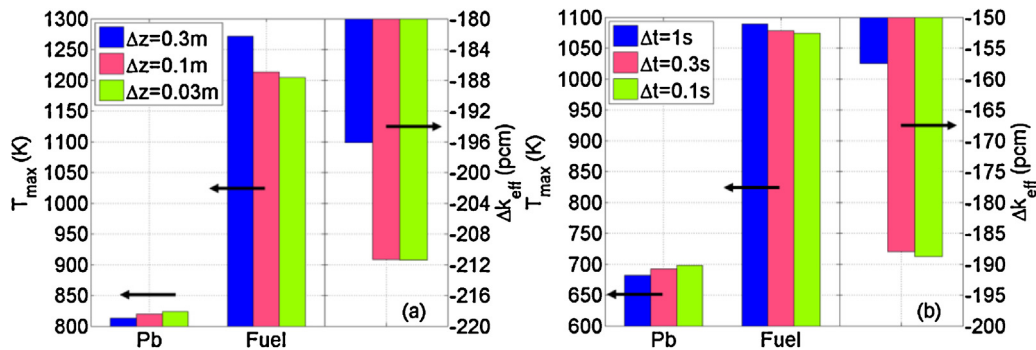


Fig. 12. Numerical convergence study for the criticality search test. (a) Space convergence: comparison between asymptotic values of the maximum temperature of Pb and fuel, and of Δk_{eff} , for three different values of the axial mesh size Δz , at fixed TH time step $\Delta t = 0.1\text{ s}$. (b) Time convergence: comparison between the values of the same three variables at time $t = 3.5\text{ s}$ in the TH module, for three different values of the TH time step Δt , at fixed $\Delta z = 0.03\text{ m}$.

evolution due to the power profile changes, while the NE module performs a steady-state computation at each TH time step Δt using in input the newly computed temperature distribution.

The behavior of Δk_{eff} computed by the NE module at each iteration is reported in Fig. 10, together with the evolution of the maximum temperature of fuel and Pb. The temperature increase due to the power input is reflected by a progressive reduction of k_{eff} , due to the negative temperature feedback described in Section 2.3. Fig. 11 clearly highlights the different spatial distributions of neutron flux and temperature. In particular, as expected, the highest neutron flux is located in the central part of the reactor (Fig. 11a), far away from the boundaries where the leakages strongly reduce the flux. This leads to the Pb temperature peaking at the outlet of the central assemblies (Fig. 11b).

Because of the severe conditions associated to the full power input in a cold configuration, if we use (8), then $T_s(z, t) < T_{\text{Pb}}(z, t)$ would result at the beginning of the transient, which makes no sense. Therefore, whenever this occurs, we force $T_s(z, t) = T_{\text{Pb}}(z, t)$, and this explains the delayed reaction of the Pb to the pin temperature increase, because the heat transfer is inhibited until the T_s computed from (8) becomes larger than T_{Pb} (see Fig. 10). These ad-hoc recipes are justified by the fact that we are not following a real transient here, but rather trying to find a sufficiently accurate, NE/TH coupled steady-state condition of the reactor core.

Fig. 12 shows the results of the numerical convergence study performed for this test problem: we consider the sensitivity of the solution to two discretization parameters, namely the (uniform) axial mesh size Δz (Fig. 12a) and the (uniform) time step Δt , both adopted by the TH module (Fig. 12b). It confirms that the computed solution can be considered reasonably independent of Δz and Δt .

4. Conclusions and perspective

The new code FRENETIC has been presented for the coupled neutronic/thermal-hydraulic analysis of lead-cooled fast nuclear reactors, as currently foreseen in the European program within the Generation IV framework.

Both the neutronic and thermal-hydraulic modules of FRENETIC have been benchmarked against analytical solutions. The results of a criticality search have also been discussed, showing the present capabilities of the code to tackle a coupled neutronic/thermal-hydraulic problem. Grid size and time step independence of the numerical solution have been verified by a computational convergence study.

The extension of the neutronic model to include 3D and transient capabilities is under way. In a separate, parallel effort, the multi-hexagon thermal-hydraulic model will be validated against experimental data. Eventually, the coupled

neutronic/thermal-hydraulic model should also be validated against experimental data.

Acknowledgments

This work was partially financially supported by the Italian Ministero per lo Sviluppo Economico, under the Ricerca di Sistema Elettrico program. We thank M. Canella for the preliminary implementation of the nodal equations in the neutronic module. We are also grateful to the two reviewers for comments that significantly improved the quality and readability of the paper.

References

- Abramowitz, M., Stegun, I.A., 1965. *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Dover, New York.
- Alemberti, A., Carlsson, J., Malambu, E., Orden, A., Struwe, D., Agostini, P., Monti, S., 2011. European lead fast reactor – ELSY. *Nucl. Eng. Des.* 241, 3470–3480.
- Alemberti, A., unpublished results. ELFR-DEMO: ALFRED system configuration. Presented at the International Workshop on Innovative Nuclear Reactors Cooled by Heavy Liquid Metals: Status and Perspectives. Pisa, April 17th–20th, 2012.
- Bonifetto, R., Dulla, S., Ravetto, P., Savoldi Richard, L., Zanino, R., 2012a. *Progress in multi-physics modeling of innovative lead-cooled fast reactors*. *Trans. Fusion Sci. Technol.* 61, 293–297.
- Bonifetto, R., Savoldi Richard, L., Zanino, R., 2012b. Validation of a 1D thermal-hydraulic module against Circe experimental data. Report RdS/2012/040. Available at http://www.enea.it/it/Ricerca_sviluppo/documenti/ricerca-di-sistema-elettrico/nuovo-nucleare-fissione/lp3/2011/040-lp3-rds-pdf
- Bortot, S., Cammi, A., Artioli, C., 2012. Dynamic performance assessment of MOX and metallic fuel core options for a Gen-IV LFR demonstrator. *Prog. Nucl. Energy* 54, 112–122.
- Carta, M., Personal communication, 2012.
- Davis, C.B., Shieh, A.S., 2000. Overview of the use of ATHENA for thermal-hydraulic analysis of systems with lead-bismuth coolant. In: *Proceedings of the 8th International Conference on Nuclear Engineering (ICONE)*, Baltimore, MD, USA, April 2–6, Available at <http://www.inl.gov/technicalpublications/Documents/3314512.pdf>
- Deffrennes, M., unpublished results. EU energy policy: context and challenges for nuclear energy. Presented at International Workshop on Innovative Nuclear Reactors Cooled by Heavy Liquid Metals: Status and Perspectives. Pisa, April 17th–20th, 2012.
- Dulla, S., Mund, E.H., Ravetto, P., 2008. The quasi-static method revisited. *Prog. Nucl. Energy* 50, 908–920.
- GIF, 2002. A Technology Roadmap for Generation IV Nuclear Energy Systems, GIF-002-00, USDOE Nuclear Energy Research Advisory Committee and the Generation IV International Forum (2002). Available at <http://www.gen-4.org/PDFs/GenIVRoadmap.pdf>
- Hursin, M., Downar, T.J., Kochunas, B., 2012. Analysis of the core power response during a PWR rod ejection transient using the PARCS nodal code and the DeCART MOC code. *Nucl. Sci. Eng.* 170, 151–167.
- <https://laws.lanl.gov/vhosts/mcnp.lanl.gov/> (accessed 24.10.12).
- <http://www.cathare.cea.fr> (accessed 24.10.12).
- <http://www.inl.gov/relap5/> (accessed 24.10.12).
- <http://www.leader-fp7.eu/> (accessed 24.10.12).
- <http://www.tlk-thermo.com/> (accessed 24.10.12).
- Incropera, F.P., Dewitt, D., 2006. *Fundamentals of Heat and Mass Transfer*, 6th ed. Wiley & Sons, New York, USA.

- Lawrence, R.D., 1983. The DIF3D nodal neutronics option for two- and three-dimensional diffusion-theory calculations in hexagonal geometry. Argonne National Laboratory Report ANL-83-1.
- Malambu, E.L., Mund, E.H., 1995. A self-consistent nodal method in response matrix formalism for the multigroup diffusion equations. *Ann. Nucl. Energy* 23, 99–118.
- Mansani, L., unpublished results. ELFR system configuration and main components. Presented at the International Workshop on Innovative Nuclear Reactors Cooled by Heavy Liquid Metals: Status and Perspectives. Pisa, April 17th–20th, 2012.
- Meloni, P., Bandini, G., Polidori, M., 2008. EFIT reactor simulation coupling neutronics/thermal-hydraulics with the RELAP5/PARCS code. In: Proceedings of the International Congress on the Advances in Nuclear Power Plants (ICAPP), Paper 8273, Anaheim, CA, USA, June 8–12, 2008. Available at <http://nuklear-server.ka.fzk.de/OFMS/Web%2FMain%2FPublications%2F2008%2FICAPP2008%2FMeloni.8273-draft.pdf>
- Mikhin, V.I., Polidori, M., Voukelatou, K., Bornia, G., Finelli, M., Manservigi, S., 2011. Development and validation of FEM-LCORE code for the thermal hydraulics of open cores. *Ricerca di Sistema Elettrico*, Report RdS/2011/2. Available at <http://www.enea.it/it/Ricerca.sviluppo/documenti/ricerca-di-sistema-elettrico/nuovo-nucleare-fissione/lp3/rds-2-lp3.pdf>
- Mikityuk, K., Pelloni, S., Coddington, P., Bubelis, E., Chawla, R., 2005. FAST: an advanced code system for fast reactor transient analysis. *Ann. Nucl. Energy* 32, 1613–1631.
- OECD-NEA, 2007. *Handbook on Lead–Bismuth Eutectic Alloy and Lead Properties, Materials, Compatibility, Thermal-Hydraulics and Technologies*. OECD-NEA.
- Popov, S.G., Carbajo, J.J., Ivanov, V.K., Yoder, G.L., 2000. Thermophysical properties of MOX and UO₂ fuels including the effects of irradiation. ORNL Report TM-2000/351.
- Rhoades, W.A., Azmy, Y.Y., 1995. Three-dimensional Sn calculations with the Oak Ridge TORT code. In: International Conference on Mathematics and Computations, Reactor Physics, and Environmental Analyses, Portland, OR, April 30–May 4, 1995. Available at <http://www.osti.gov/bridge/servlets/purl/10114812-NsMjv0/webviewable/10114812.pdf>
- Rimpault, G., et al., 2002. The ERANOS code and data system for fast reactor neutronic analyses. In: *Proceedings of the International Conference PHYSOR2002*, Seoul, Korea.
- Sanchez, R., et al., 2010. APOLLO2 year 2010. *Nucl. Eng. Technol.* 42, 474–499.
- Savoldi Richard, L., Casella, F., Fiori, B., Zanino, R., 2010. The 4C code for the cryogenic circuit conductor and coil modeling in ITER. *Cryogenics* 50, 167–176.
- Savoldi, L., Zanino, R., 2000. M&M: multi-conductor Mithrandir code for the simulation of thermal-hydraulic transients in superconducting magnets. *Cryogenics* 40, 179–189.
- Sobolev, V., Malambu, E., Ait Abderrahim, H., 2007. Definition of the ELSY Core Configuration Parameters: Preliminary Specifications for the ELSY-600 Core. Deliverable M7, EURATOM FP6 ELSY Project, Contract N° FI6W-2006-036439.
- Todreas, N.E., Kazimi, M.S., 1993. *Nuclear Systems*, vol.1. Hemisphere, New York.
- Yamano, H., Fujita, S., Tobita, Y., Sato, I., Niwa, H., 2008. Development of a three-dimensional CDA analysis code: SIMMER-IV and its first application to reactor case. *Nucl. Eng. Des.* 238, 66–73.
- Zanino, R., Bonifetto, R., Ciampichetti, A., Di Piazza, I., Savoldi Richard, I., Tarantino, M., 2012. Validation of the thermal-hydraulic model in the FRENETIC code against data from the ENEA-CIRCE experiment. *Trans. Am. Nucl. Soc.* 107, 1395–1398.